

Sri Lanka Institute of Information Technology

EM 1 - LAB03



STD Number – IT26101833

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Exercise Command:

The image displays two screenshots of the Octave IDE, showing the execution of commands to find the roots of a complex number and plot them.

Top Screenshot: The Octave IDE window shows the current directory as `C:\Users\T26101833`. The workspace contains variables `ang0`, `ang1`, `ang2`, `ang3`, `p0`, `p1`, `p2`, `p3`, `r0`, and `r1`. The command history shows the execution of the `clear` command.

```
1 clear;
2 close;
3 clc;
4
5 p0 = -8i;
6
7 r0 = abs(p0); %modulus of p0
8
9 p1 = r0*(1/3);
10 p2 = p3 - p1;
11
12 r1 = abs(p1);
13 r2 = abs(p2);
14 r3 = abs(p3);
15
16 ang0 = arg(p0);
17
18 ang1 = (ang0 / 3);
19 ang2 = ang1 + (2*pi)/3;
20 ang3 = ang1 + 2*(2*pi)/3;
21
22
23 x1 = r1*cos(ang1); y1 = r1*sin(ang1);
24 x2 = r2*cos(ang2); y2 = r2*sin(ang2);
25 x3 = r3*cos(ang3); y3 = r3*sin(ang3);
26
27 %drawing the plot
28 figure
29 plot([0, x1], [0, y1], [0, x2], [0, y2], [0, x3], [0, y3]);
30 xlabel('Real');
31 ylabel('Imag');
32 title('Roots of complex number -8i')
33 grid('on');
34 axis([-3, 3, -2.5, 2.5]); %set limit to x axis and y axis
35 text(x1,y1,['(" num2str(r1) ", " num2str(ang1) ")'], "fontsize", 15) % text label for (r1, angl)
36 text(x2,y2,['(" num2str(r2) ", " num2str(ang2) ")'], "fontsize", 15) % text label for (r2, angl)
37 text(x3,y3,['(" num2str(r3) ", " num2str(ang3) ")'], "fontsize", 15) % text label for (r, angl)
38
39
40
```

Bottom Screenshot: The Octave IDE window shows the current directory as `C:\Users\T26101833`. The workspace contains variables `ang0`, `ang1`, `ang2`, `ang3`, `p0`, `p1`, `p2`, `p3`, `r0`, and `r1`. The command history shows the execution of the `clear` command.

```
13 r2 = abs(p2);
14 r3 = abs(p3);
15
16 ang0 = arg(p0);
17
18 ang1 = (ang0 / 3);
19 ang2 = ang1 + (2*pi)/3;
20 ang3 = ang1 + 2*(2*pi)/3;
21
22
23 x1 = r1*cos(ang1); y1 = r1*sin(ang1);
24 x2 = r2*cos(ang2); y2 = r2*sin(ang2);
25 x3 = r3*cos(ang3); y3 = r3*sin(ang3);
26
27 %drawing the plot
28 figure
29 plot([0, x1], [0, y1], [0, x2], [0, y2], [0, x3], [0, y3]);
30 xlabel('Real');
31 ylabel('Imag');
32 title('Roots of complex number -8i')
33 grid('on');
34 axis([-3, 3, -2.5, 2.5]); %set limit to x axis and y axis
35 text(x1,y1,['(" num2str(r1) ", " num2str(ang1) ")'], "fontsize", 15) % text label for (r1, angl)
36 text(x2,y2,['(" num2str(r2) ", " num2str(ang2) ")'], "fontsize", 15) % text label for (r2, angl)
37 text(x3,y3,['(" num2str(r3) ", " num2str(ang3) ")'], "fontsize", 15) % text label for (r, angl)
38
39
40
41 %Optional - drawing the circle of radius pi
42 xx = linspace(-pi, pi, 50);
43 yy1 = sqrt(pi^2 -xx.^2);
44 yy2 = -1*sqrt(pi^2 -xx.^2);
45 hold('on'); %to display the next plot in the same window
46 plot(xx,yy1,'linestyle','--','k',xx,yy2,'linestyle','--','k');
47
48
49
50
51
```

Exercise Output:

